
Abduction: a reasoning axis where learned simulation and unaided frontier LLMs dissociate

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Abstract

We study *abduction*, the recovery of an unobserved initial condition from a noisy trajectory, as a distinct axis of reasoning that sits alongside forward execution and goal-directed control. We compare frontier LLMs, including the dedicated reasoning model o3, against a deliberately non-transformer reasoner. That reasoner learns a world model from black-box (state, [action,] next-state) transitions, never reading the true rule, and reasons by simulating the model it learned. The inference is **simulation-based inference** [8]: maximum-a-posteriori (MAP) over forward rollouts. Our contribution is the capability dissociation it exposes, not the method. A reasoning model closes forward execution and interactive control; o3 solves Sokoban control 17/18, close to our 18/18. It does not close abduction. As the trajectory grows the LLMs stay near the single-noisy-frame floor and collapse at long horizon, while the learned-simulation reasoner tracks the true-dynamics recoverability oracle. At cellular-automaton (CA) horizon 18, ours recovers 30/30 [0.88, 1.00], against o3 at 1/30 [0.01, 0.17] (4/30 at high effort) and Sonnet at 0/30 [0.00, 0.11], and the gap widens with difficulty. A second dedicated reasoning model from a different lab (Claude Opus 4.8 in adaptive thinking) replicates the pattern in a smaller probe: it closes forward and control yet does not reliably abduct. The failure is one of *unaided* inference: given code execution the same model writes the search and closes the gap (20/30, matching ours), which points to in-context exact inverse search, rather than reasoning in general, as the bottleneck. The learned model is load-bearing: recovery follows its fidelity and falls to the floor when we degrade it, and the reasoner equals the oracle. This is a forward/control-pass, inverse-fail *dissociation* (o3 passes the axes it shares with our reasoner and fails only the inverse one), and it holds across two structurally different worlds, a reversible 1-D cellular automaton and an irreversible 2-D Sokoban.

1 Introduction

To reconstruct how a fire started, you reason backward from the wreckage to the spark. To predict where a thrown ball will land, you run the world forward. These are different acts, and a mind can be sharp at one and clumsy at the other. We show that frontier language models are lopsided in exactly this way: they run a rule forward, and they intervene to steer it, but they cannot reliably run it *backward* to recover where a noisy trajectory began.

Most evaluations of reasoning in large language models hide this, because they collapse reasoning into a single scalar on which one model is simply “better at reasoning” than another. We argue instead that reasoning has distinguishable *axes*, and that current frontier models pass some while failing others in a way a scalar score conceals. Figure 1 summarizes the result.

The core, after we cede the inference *method* to simulation-based inference (SBI) [8] and the bare observation that “LLMs fail backward” to the reversal-curse literature [2], is a conjunction that,

*Code and data: <https://github.com/integrallis/abduction-dissociation>

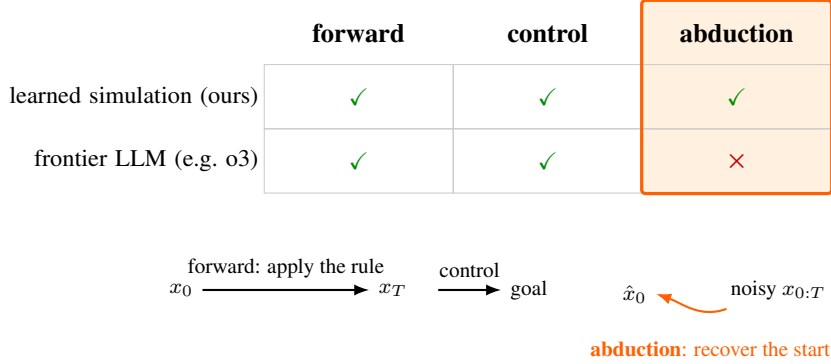


Figure 1: **The dissociation.** A reasoner can run a dynamical world *forward*, *control* it toward a target, or *abduct* the initial condition from a noisy trajectory (the inverse). A frontier reasoning model (o3) closes the two forward-facing axes it shares with our learned-simulation reasoner, but not the inverse one. Our reasoner closes all three, sitting at the true-dynamics recoverability oracle. The orange column is the abduction gap.

to our knowledge, no prior work draws. In plain terms: a small non-transformer model learns the world’s rule just by watching transitions, never being shown it, and recovers a hidden starting state by simulating that learned rule forward and keeping the candidate whose trajectory best matches the observation. Put to the same backward task, unaided frontier LLMs pass forward execution and control yet fail the recovery; even a dedicated reasoning model (o3) fails it, the gap grows with difficulty, and it holds in two structurally opposite worlds. Formally:

a non-transformer reasoner whose forward rule is learned from black-box transitions does initial-condition recovery by MAP-over-learned-rollouts, and this exposes a **forward/control-pass, inverse-fail dissociation** in *unaided* frontier LLMs that **survives a dedicated reasoning model (o3)**, with the gap **widening with difficulty**, shown **unchanged across a reversible 1-D CA and an irreversible 2-D Sokoban**.

The conjunction is the contribution, and each element sets it beyond a nearby prior result: the simulator is *learned* from black-box transitions rather than given (beyond standard SBI); the dissociation *survives* a dedicated reasoning model (beyond the reversal curse); it holds *across* a reversible CA and an irreversible Sokoban (beyond a single domain); and the controls show the learned model carries the inference, not a leaked oracle.

Scope, stated up front. The comparison is deliberately asymmetric, and we surface that here rather than only in Section 8: our inverter is a *hand-structured* simulator family (a weight-shared local rule matched to each substrate, with exhaustive candidate search over small, enumerable state spaces under a *known* symmetric noise model), set against frontier LLMs *forbidden from writing or executing code*. The precise, well-supported reading is therefore not that abduction is a new cognitive faculty, but that *within this benchmark family* exact initial-state recovery behaves as a *separable* reasoning dimension on which unaided frontier LLMs fail while a structured learned simulator succeeds up to a measured tractability wall. The gap is a property of *current unaided models as evaluated*, not a claim of architectural impossibility.

The three axes. We separate three things a reasoner can be asked to do over a dynamical world (Table 1). *Forward execution* applies the rule N steps. *Control*, or planning, finds $\leq B$ interventions that force a target state. *Abduction* recovers the initial condition from a noisy trajectory; it is the form of fault diagnosis, root-cause analysis, forensic reconstruction, and scientific parameter inference, which all infer an unobserved cause from its noisy consequences. The first two are gaps that separate weak models from strong ones: o3 holds forward execution to $N=16$ where weak models fail by $N=4$, and it closes control (Sokoban 17/18, against our 18/18 and Sonnet’s 10/18). **Abduction is the axis o3 does not close** (CA horizon-18: 1/30; Sokoban 5/8 $\rightarrow$ 2/8), while our reasoner sits at the oracle ceiling. We measure that contrast on the *same* worlds, with control as the matched

Table 1: The three axes of reasoning. Forward and control are *weak-LLM* gaps that a dedicated reasoning model closes; abduction is the one it does not. The contrast is measured on the same worlds, with control as the matched within-domain comparison.

axis	what it tests	frontier-LLM behavior	our reasoner
forward execution	apply the rule N steps	o3 exact to $N=16$; weak LLMs fail by $N=4$	exact
control / planning	find $\leq B$ interventions to force a target	o3 closes it (Sokoban 17/18)	18/18
abduction	recover the initial condition from a noisy trajectory	o3 does not close it (CA H=18: 1/30; Sokoban $n=8$: 5/8 $\rightarrow$ 2/8)	oracle ceiling

within-domain comparison, and that contrast is the contribution. o3 closes every axis it shares with our reasoner, except the inverse one.

2 The playgrounds

We use two worlds chosen to be structurally opposite, so that a shared result cannot be a single-domain artifact (Figure 2).

A reversible 1-D cellular automaton. The first world is a ring of L binary cells that updates in discrete time. An *elementary* cellular automaton sets each cell’s next value from its three-cell neighborhood (left, self, right) through a fixed Boolean lookup table; the eight possible neighborhoods give $2^8 = 256$ such tables, and Wolfram’s convention names each one by reading its eight outputs as an integer, hence “Rule 110” and “Rule 90”. We use a *second-order* lift, $\text{next}_i = f(\text{cur}_{i-1}, \text{cur}_i, \text{cur}_{i+1}) \oplus \text{prev}_i$, which is *reversible*: given (cur, next) one recovers prev, so trajectories are well-posed in both directions. A state is the pair (prev, cur); an instance rolls a hidden initial condition forward T steps and reveals the trajectory under per-cell bit-flip noise.

The two rules matter because of **computational irreducibility**. *Rule 110* is class-4 and Turing-complete: there is no closed form for its configuration at step T , so the only way to know it is to run all T steps, and inverting it therefore requires simulation. *Rule 90* is additive ($f(a, b, c) = a \oplus c$): it is linear, its evolution superposes, and its step- T state has a closed-form shortcut, visible as the regular Sierpinski triangle in Figure 2a. Rule 90 is our control. A method that “solved” the inverse only by exploiting a shortcut would behave the same on both rules; a method that truly simulates should behave differently, and ours does.

An irreversible 2-D Sokoban. The second world is Sokoban: an agent (the pusher) moves on a walled grid and shoves boxes onto goal cells, one box per push, never pulling. Pushes are *irreversible*: a box driven against a wall cannot be undone. Unlike the CA, information about the past is genuinely destroyed. A state is the grid layout (walls, boxes, pusher); an instance rolls a hidden initial layout forward by a sequence of moves and reveals it under noise. Scoring in both worlds is exact: a recovered initial condition counts only if every bit, or every cell, matches.

3 The reasoner

The reasoner never reads the rule. It learns a forward model from black-box transitions, then inverts that model by simulation. This is **simulation-based inference** (SBI) [8]: recover a latent input by rolling a forward simulator and scoring against the observation. The one thing we add is that the simulator is *learned from black-box transitions*, with no rule access; the other is the head-to-head against LLMs across the three axes. Figure 3 and Algorithm 1 give the mechanism in full.

The learned forward model. The model is a single **weight-shared local rule** f_θ : a small MLP that maps one cell’s neighborhood ($\text{prev}_i, \text{cur}_{i-1}, \text{cur}_i, \text{cur}_{i+1}$) to that cell’s next value, applied identically at every cell and every ring width. Because no width parameter appears in the weights, a model fit on small rings instantiates unchanged on larger ones, so size-generalization is structural rather than

(a) Reversible 1-D cellular automaton (space $\rightarrow$, time $\downarrow$)

(b) Sokoban

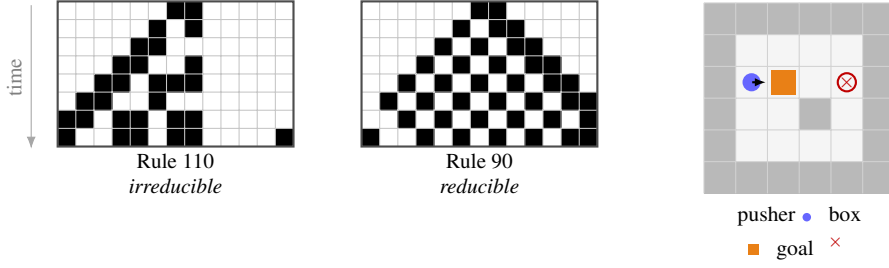


Figure 2: **The two playgrounds.** (a) Real space-time traces of the second-order CA from one seed (width-13 ring shown; the rule is width-independent). Each row is one configuration; time runs down. *Rule 110* (left) is computationally *irreducible* (class-4, Turing-complete): its irregular pattern cannot be predicted faster than by simulating every step. *Rule 90* (right) is *reducible* (additive, $f(a, b, c) = a \oplus c$): the regular Sierpinski structure means a closed-form shortcut exists, so it is our control. (b) A Sokoban instance: a pusher moves on a walled grid and shoves boxes onto goals. Pushes are *irreversible* (a box shoved against a wall cannot be pulled back), so this world is structurally unlike the reversible CA.

Algorithm 1 MAP abduction over the learned model

Require: noisy trajectory $\tilde{y}_{0:T}$; learned rule f_θ ; candidate set $\mathcal{X}_0$

- 1: $\text{best} \leftarrow -\infty$; $\hat{x}_0 \leftarrow \emptyset$
 - 2: **for** each candidate $x_0 \in \mathcal{X}_0$ **do**
 - 3: $\hat{y}_{0:T} \leftarrow \text{ROLLOUT}(f_\theta, x_0, T)$ $\triangleright$ simulate the *learned* model, never the true one
 - 4: $\ell \leftarrow \sum_{t=0}^T \sum_i \log p(\hat{y}_{t,i} \mid \tilde{y}_{t,i})$ $\triangleright$ whole-trajectory per-cell log-likelihood
 - 5: **if** $\ell > \text{best}$ **then** $\text{best} \leftarrow \ell$; $\hat{x}_0 \leftarrow x_0$
 - 6: **return** $\hat{x}_0$
-

hoped for. The model reads neither the rule code nor the width; the training details (architecture, optimizer, data) are in Section 4.

Abduction as MAP over learned rollouts. Given a noisy trajectory, the reasoner enumerates candidate initial conditions, rolls each one forward T steps under f_θ , scores the simulated trajectory against the observation by a per-cell likelihood, and returns the maximum-a-posteriori candidate (Algorithm 1). Evidence is aggregated across the *whole* trajectory, and that aggregation is the inference. The inverse problem is not hard once the rule is known; it is classical pre-image computation. The contribution is the learned substrate and the dissociation it exposes.

4 Experimental setup

Instances and the noise channel. A CA instance draws a hidden initial condition $x_0 = (\text{prev}_0, \text{cur}_0)$ uniformly from $\{0, 1\}^L \times \{0, 1\}^L$, rolls it forward H steps under the true rule (the horizon H is the T of Algorithm 1), and reveals all $H+1$ frames through an i.i.d. bit-flip channel: every cell is flipped independently with probability η (we use $\eta = 0.2$). Sokoban noises only the box and agent cells, leaving walls and goals intact, so the static board is read from the first frame. Unless stated otherwise the CA runs at $L = 8$, $H \in \{6, 12, 18\}$; size generalization is tested at $L \in \{11, 12\}$, both outside the $L \leq 10$ training range. Scoring is exact: a recovered initial condition (IC) counts only if it matches the truth in every bit (CA) or cell (Sokoban).

Likelihood, prior, candidate set, and oracle. Under the symmetric channel the per-cell likelihood is $p(\tilde{y} \mid \hat{y}) = (1 - \eta)^{[\tilde{y}=\hat{y}]} \eta^{[\tilde{y} \neq \hat{y}]}$, so with a uniform prior over $\mathcal{X}_0$ the MAP of Algorithm 1 reduces to the candidate whose rollout agrees with the observation in the most cells. The candidate set $\mathcal{X}_0$ is, for the CA, all 2^{2L} pairs (prev, cur) (enumerated exhaustively; feasible to $L \approx 12$, Section 8); for

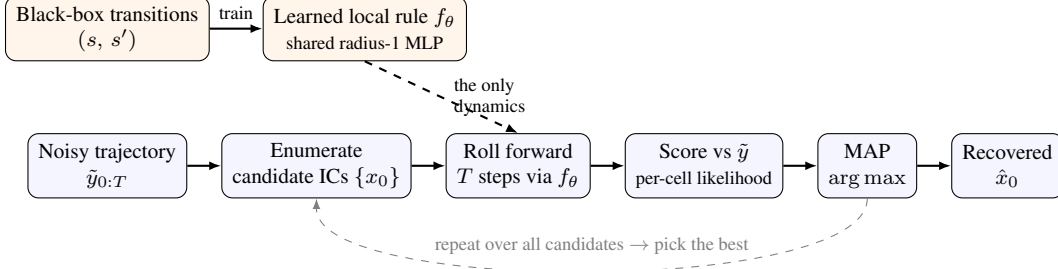


Figure 3: **The reasoner.** A forward model is *learned* from black-box (state, [action,] next-state) transitions (top): one weight-shared radius-1 rule f_θ , applied at every cell and every system size, so it generalizes across width by construction. Abduction (bottom) is maximum-a-posteriori inference over that learned model: enumerate candidate initial conditions, roll each one forward under f_θ , score the simulated trajectory against the noisy observation, and return the best (Algorithm 1). The true environment is never queried during the search; f_θ supplies all dynamics.

Sokoban, the $\binom{\text{free}}{k}$ (free $- k$) placements of the $k=2$ boxes and the pusher over free cells, with walls and goals held to their observed positions. Ties take the lowest index. The **recoverability oracle** is Algorithm 1 run with the *true* rule in place of f_θ (same $\mathcal{X}_0$, likelihood, and noise draw); it is the best exact recovery the noise allows, so “ours = oracle” means the learned model loses nothing to the true one.

The learned model. f_θ is a two-layer MLP ($4 \rightarrow 16 \rightarrow 1$, ReLU), trained by binary cross-entropy (Adam, learning rate 0.01, 1500 steps, batch 512) on black-box (state, next-state) transitions pooled over widths $\{6, 7, 8, 9, 10\}$ (400 per width); the predicted bit is $\text{logit} > 0$. *Fidelity* (the degrade control, Section 6) is its held-out one-step whole-configuration exact-match rate, dialed by the training-step budget. *Exact joint recovery* (the main metric) is the per-instance indicator that the recovered $\hat{x}_0$ matches the true x_0 in every bit.

LLM baselines and the evaluation protocol. We compare a dedicated reasoning model (o3, reasoning_effort=medium, with a high-effort robustness run), a *second* dedicated reasoning model from a different lab (Claude Opus 4.8 in adaptive/extended thinking, a smaller $n = 5/\text{cell}$ probe), a frontier chat model (Claude Sonnet 4.6), and, on the forward axis, a deliberately weak model (GPT-4o-mini). Decoding is temperature 0, up to 8192 tokens for the o-series and chat models. For the adaptive-thinking model the chain-of-thought counts against the output cap, so we raise it to the 128k ceiling and log stop_reason; an instance whose thinking overruns even 128k is re-run at lower effort, never scored on a truncated trace. Each model is handed the *exact rule table*, the dynamics equation, and the noise level in-prompt and may use a scratchpad; tools and code execution are disallowed, since writing the inversion program is retrieval of a known algorithm, not abduction. Answers must end in fixed tags (e.g. FINAL_X0 / FINAL_X1); extraction is strict and we report the **parse rate**, so a malformed answer is never scored as a reasoning failure (Appendix A). Sample sizes: CA ours $n=200/\text{rule}$ over 5 seeds, CA LLMs $n=15/\text{rule}/\text{horizon}$; Sokoban ours $n=36/\text{size}$ over 3 seeds, LLMs $n=8$. Code and the cached LLM transcripts are released with the paper.

5 Results

5.1 Cellular automaton: the abduction gap, and it widens

Figure 4 shows one instance: from a trajectory with 12 of its 72 bits flipped, the reasoner recovers the hidden IC exactly. In aggregate, our learned-model MAP is pinned to the true-dynamics recoverability oracle at every difficulty, across 5 seeds, $n = 200/\text{rule}$. Ours equals the true-dynamics MAP at every (rule, noise), and ours falls only where the *oracle ceiling* itself falls, to task ambiguity, never below it. The reasoner is therefore not under-performing. Where exact recovery drops, the ceiling drops with it.

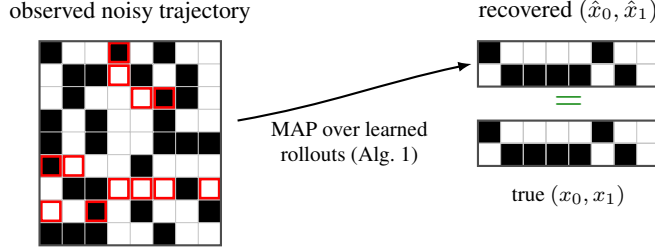


Figure 4: **A worked instance.** A hidden Rule-110 initial condition is rolled forward $H = 8$ steps and revealed through the bit-flip channel, which flips **12 of the 72 bits** (red outlines). From this noisy trajectory alone, the reasoner’s MAP over learned rollouts (Algorithm 1) recovers the true first two rows (x_0, x_1) *exactly*: the recovered pair (upper right) equals the truth (lower right). $L = 8$, noise 0.2, seed 2.

Table 2: CA abduction *per rule* (exact joint recovery, $n = 15/\text{cell}$). Ours rises with horizon on both rules while the LLMs stay near zero on both, so the gap is not a rule artifact. At $H = 12$ ours is higher on irreducible Rule 110 (13/15) than reducible Rule 90 (9/15), consistent with the inverted wall (Appendix B; the rules tie at $H=6$ and $H=18$): the irreducible rule’s wider identifiability gap should make the inverse *easier*.

H	rule	ours	o3	Sonnet
6	Rule 110	7/15	2/15	0/15
6	Rule 90	7/15	1/15	0/15
12	Rule 110	13/15	4/15	1/15
12	Rule 90	9/15	1/15	0/15
18	Rule 110	15/15	1/15	0/15
18	Rule 90	15/15	0/15	0/15

This recovery generalizes across width. Trained only on $L \leq 10$, the same rule abducts at unseen widths $L = 11$ (57/60 and 55/60 for Rules 110 and 90) and $L = 12$ (58/60, 49/60), matching the oracle at every cell: size generalization, not in-distribution reuse.

The gap to the LLMs grows with difficulty rather than shrinking: it *widens with trajectory length* ($n = 15/\text{rule}$ across horizons $H \in \{6, 12, 18\}$, transcripts cached). Figure 5 plots exact joint recovery, pooled over both rules ($n = 30/\text{cell}$). At $H = 6$, ours is 0.467, o3 0.100, Sonnet 0.000; at $H = 12$, ours 0.733, o3 0.167, Sonnet 0.033; at $H = 18$, ours 1.000, o3 0.033, Sonnet 0.000. Pooled at $H = 18$, ours recovers 30/30 [0.88, 1.00] against o3 at 1/30 [0.01, 0.17] and Sonnet at 0/30 [0.00, 0.11], with non-overlapping Wilson 95% intervals. Because the design reuses seeded instances, the comparison is *paired*: an exact McNemar test over all matched instances rejects equality decisively (ours vs o3 $p = 1.4 \times 10^{-17}$, vs Sonnet $p = 5.4 \times 10^{-20}$; ours vs Opus 4.8 $p < 10^{-3}$), and ours never loses a discordant instance to any LLM ($c = 0$ throughout; experiments/significance.py). The gap is not an effort artifact: at reasoning_effort=high o3 still reaches only 4/30 (parse rate 27/30, so these are reasoning, not formatting, failures), and all four of its recoveries fall on the *reducible* Rule 90 (4/15); on the irreducible Rule 110 it stays at 0/15, making headway only where a shortcut exists. **Ours rises**, because the MAP gains constraints as the trajectory lengthens and tracks the rising recoverability ceiling, while the **LLMs decay**, because more noisy frames have to be hand-denoised. o3 is not uniformly zero. It recovers a few *short* trajectories at $H = 6$ to 12 before collapsing to near zero at $H = 18$. That is the claim: o3 does not *reliably* abduct, and it decays with difficulty.

Graded recovery, not just exact-match. Exact whole-IC scoring could understate the LLMs if their misses were near-misses, so we re-score the same cached transcripts by Hamming distance to the true IC (out of $2L=16$ bits; random-IC chance is 8). They are not near-misses. The LLMs sit ≈ 3 to 4 bits wrong at every horizon (o3 4.1/2.7/4.3, Sonnet 4.1/3.0/3.9 at $H=6/12/18$), about halfway between exact recovery and chance, with at most half of answers within two bits. Their error does not fall as the problem becomes easier to invert: at $H=18$ the exact enumerated posterior places 0.99 of its mass on the true IC (the trajectory nearly determines it) and our learned-MAP

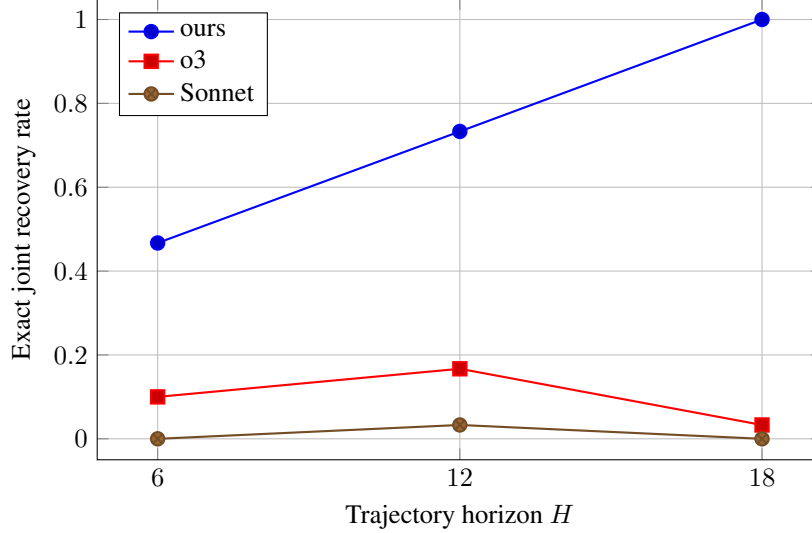


Figure 5: The CA abduction gap widens with trajectory length: ours rises while the LLMs degrade. Pooled over both rules, $n = 30$ per cell; exact joint recovery rate.

recovers it exactly (mean Hamming 0, the true IC the unique mode, mean rank 0), yet the LLMs stay ≈ 4 bits off. The oracle posterior mass rises $0.34 \rightarrow 0.75 \rightarrow 0.99$ with horizon and ours tracks it ($1.70 \rightarrow 0.87 \rightarrow 0.00$ mean Hamming), while the LLMs are flat. The graded view sharpens the dissociation rather than softening it: the recoverable signal is present and our reasoner extracts all of it as it appears, while the LLMs extract only part and do not improve as identifiability rises (experiments/ca_graded_metric.py; a re-score of the cached transcripts, no new calls).

A second reasoning model, from a different lab. The class claim should not rest on one model. We replicate the head-to-head with Claude Opus 4.8 in adaptive (extended) thinking, a dedicated reasoning model from a different vendor than o3, as a smaller probe ($n = 5/\text{cell}$, same seeded instances). It *closes the forward axis*, with exact whole-configuration recovery 12/12 at $N \in \{16, 32\}$ on both rules, holding even at $N=32$ where o3 cracks. On abduction, though, it does not rise with horizon the way the learned MAP does. Pooled it recovers 10/30 against ours at 21/30 on the same instances, and the ours-minus-Opus gap *widens* from 0 at $H=6$ (it ties ours on both rules) to +4 on irreducible Rule 110 and +2 on Rule 90 at $H=18$ (per-rule Opus recovery: Rule 110 3/1/1, Rule 90 1/1/3 at $H=6/12/18$). Opus recovers a larger fraction than medium-effort o3, so the dissociation is *weaker* for it, but the signature is the same: forward and control close while inverse recovery stalls and the gap grows with trajectory length. (All 30 calls terminated and were scored; 2 whose adaptive thinking overran the 128k budget were re-run at lower effort, the rest at medium.)

5.2 Sokoban: scaling, and the enumeration wall

The Sokoban arm scales to $n = 36/\text{size}$ across boards $\{6, 7, 8\}$ with Wilson CIs (Table 3). Ours equals the true-dynamics oracle at every size (30/36, 33/36, 30/36, identical counts). The same shared local rule applies at every board size, width-agnostic by construction, and within the enumerable range $\{6, 7, 8\}$ it sits at the recoverability ceiling (≈ 0.83 to 0.92). It stays below 1.0 only because noise plus irreversible pushes make some instances genuinely ambiguous, a property of the *task* rather than the reasoner, since ours equals oracle. The load-bearing property holds at scale: the raw-first-frame floor is near zero (1 to 4 of 36), and a degraded model collapses toward it (5 to 10 of 36, far below ours). The LLMs decay with board size (Sonnet $2/8 \rightarrow 2/8 \rightarrow 0/8$; o3 $5/8 \rightarrow 2/8$ on the existing slice), while ours holds at the ceiling.

The MAP enumeration is a *measured* wall. With $k = 2$ boxes, the candidate count $\binom{\text{free}}{k} \cdot (\text{free} - k)$ exceeds the 20,000 enumeration cap past about 8×8 : 8×8 gives 14,880 (it fits), 10×10 gives 97,527, 12×12 gives 376,740, and 16×16 gives 2,819,787, all past the cap, so exhaustive MAP

Table 3: Sokoban abduction, scaled to $n = 36/\text{size}$ across boards $\{6, 7, 8\}$ ($n = 12/\text{seed} \times 3$ seeds), with the enumeration wall measured below. Ours equals the true-dynamics oracle at every size; the raw-frame floor and the degraded model collapse toward zero; the LLM slice ($n = 8$) degrades with board size. Past about 8×8 the full candidate count $\binom{\text{free}}{k} \cdot (\text{free} - k)$ (with $k = 2$ boxes) exceeds the 20,000 cap, so MAP becomes infeasible and amortized inference is the next step. (The two 8×8 candidate counts differ by design: the scaled head-to-head rows use sparse-wall random grids, about 17,952 candidates, whereas the wall row is a combinatorial projection on a representative 32-free-cell board, 14,880; both straddle the cap, placing 8×8 exactly at the boundary.)

size	candidates	ours (== oracle)	raw floor	degraded	Sonnet ($n = 8$)
6×6	1,680	30/36 [0.68, 0.92]	4/36	10/36	2/8
7×7	6,072	33/36 [0.78, 0.97]	2/36	9/36	2/8
8×8	17,952	30/36 [0.68, 0.92]	1/36	5/36	0/8
<i>Enumeration wall</i> (candidates $\binom{\text{free}}{k} \cdot (\text{free} - k)$, cap 20,000):					
8×8	14,880	fits			
10×10	97,527	past cap $\rightarrow$ MAP infeasible			
12×12	376,740	past cap $\rightarrow$ MAP infeasible			
16×16	2,819,787	past cap $\rightarrow$ MAP infeasible			

Table 4: Sokoban control (solve = the plan reaches the goal in the true environment), $n = 18$ binned by optimal plan length. Our learned-model planner ties the classical A* true-model reference; o3 *closes* this axis; the random-shooting floor and Sonnet trail. A* and ours both touch the true model only to *verify*; the LLMs are given the rules in-prompt with code execution off. (experiments/sok_control_astar.py, sok_control_llm.py.)

	ours (learned planner)	A* (true model)	o3	Sonnet	random shooting
solved	18/18	18/18	17/18	10/18	10/18

can no longer reach the true layout. This makes the next step concrete: amortized, learned-proposal inference, which we do not build here.

5.3 Forward execution: weak models fail, o3 holds

The forward axis confirms abduction is the *specific* failure, not weakness in general. Applying the rule N steps (rule table given, scratchpad allowed, no tools), our learned rule is exact at every N . o3 is exact through $N = 16$ on both rules and cracks only at $N = 32$ (3/5 on irreducible Rule 110, 5/5 on Rule 90), while the weak model (GPT-4o-mini) collapses immediately (0/5 from $N = 4$, at chance bit-accuracy). Claude Opus 4.8 (adaptive thinking) likewise closes it, with exact 12/12 at $N \in \{16, 32\}$ on both rules, holding even at $N=32$ where o3 cracks. Forward execution separates weak models from strong ones, and both reasoning models pass it.

5.4 Within-domain control: the matched comparison

The control axis is the matched within-domain comparison that pins the claim to abduction specifically. On Sokoban control, o3 solves 17/18 against our 18/18 (Sonnet 10/18): the reasoning model *closes* this axis (Table 4). Because o3 closes control on the very world where it fails abduction, the failure cannot be waved away as a generic weakness or a formatting artifact. It is specific to the inverse direction. We include the classical reference in the table rather than only in code: A* over the *true* model solves 18/18 by construction (an instance is admitted only if true-model search finds a plan), so this is the upper reference the matched comparison sits under, not a “beats A*” claim; the contribution is the abduction axis, on which A* has no advantage to offer an LLM that cannot run it.

Is the gap just in-context search? A tool-augmented baseline says yes, and that localizes it. A natural reading is that abduction merely demands more candidate search than an *unaided* LLM can carry by hand: an in-context exact-search limit rather than a generic reasoning one. We test it directly: allowed to *write and run Python* (Anthropic’s server-side code execution), the same Opus 4.8 writes

Table 5: Load-bearing controls. All three pass. Oracle-matching shows ours is not under-performing; degrade-to-chance shows recovery tracks the learned model’s fidelity and collapses to the floor; the call-site audit shows the inverse runs entirely inside the learned model.

control	result
oracle-matching	ours == true-dynamics MAP at every (rule, noise), $n = 200/\text{rule}$ (CA) and 30/36, 33/36, 30/36 (Sokoban)
degrade-to-chance	fidelity 0.000 $\rightarrow$ recovery 0.000; 0.030 $\rightarrow$ 0.017; 0.280 $\rightarrow$ 0.033; 1.000 $\rightarrow$ 0.950; raw-frame floor 0.05; random-IC chance $\approx 1.5 \times 10^{-5}$
call-site audit	learned-rule ticks = 11 ($= H - 1$); true-CA ticks = 0 during inference

the enumerate-and-rollout MAP and closes the gap, recovering 20/30 and matching ours at 21/30 on the same instances (code ran on all 30; `experiments/ca_abduction_tool.py`), against 10/30 tool-free. This matches a broad program-aided-reasoning finding, that delegating computation to executable code substantially outperforms in-context chain-of-thought on exact symbolic tasks [10, 6]. We read this single tool-enabled slice as evidence that a major bottleneck is *in-context exact inverse search* rather than reasoning at large, not as a full account of every abductive miss; localizing it this way sharpens the claim rather than retreating from it. It is search *size*, not search-versus-none: o3 carries $N=16$ serial forward steps and closes bounded control, but the pre-image set grows with horizon. Abduction is the axis that demands pre-image search an unaided model cannot hold in-head; a learned simulator does it natively, exhaustively, verifiably, at the oracle ceiling and across system size. That is when, lacking trusted code execution or needing a checkable answer, you reach past the language model and simulate.

6 Load-bearing controls

These controls establish that the *learned model* does the reasoning, not a lookup, a hidden solver, or a leaked oracle. All three pass (Table 5).

Oracle-matching. Ours equals the true-dynamics MAP at every (rule, noise), $n = 200/\text{rule}$ on the CA, and 30/36, 33/36, 30/36 on Sokoban, identical to the oracle. So the recoverability *ceiling* bounds ours, not a deficiency in the reasoner; where ours falls, the oracle falls identically.

Degrade-to-chance. Trained at increasing fidelity, abduction exact-recovery rises monotonically from the floor to the ceiling (Figure 6; Rule 110, $L = 8$, $H = 12$, noise 0.2). Fidelity 0.000 gives recovery 0.000; 0.030 gives 0.017; 0.280 gives 0.033; 1.000 gives 0.950. The raw-1-frame floor (noisy x_0, x_1 as the estimate, no model) is 0.05, and random-IC chance is about 1.5×10^{-5} . A degraded model produces garbage candidate rollouts, so the MAP collapses to the floor; only as fidelity approaches 1 does recovery reach ≈ 0.95 . The learned model is load-bearing for abduction, in a graded manner, not a lookup table or a classical solver in disguise.

Call-site audit. We instrument the inference path, with the instance generated *before* instrumentation. The learned-rule ticks = 11 ($= H - 1$), while true-CA ticks = 0 during inference. The abduction happens entirely inside the learned model; the true environment is touched only to generate the instance and to score the recovered IC. This is the “no leaked dynamics” guarantee, established for the inverse problem.

7 Related work and positioning

Simulation-based inference. Recovering an unobserved IC or parameters by rolling a forward simulator and taking the MAP or posterior is the defining move of simulation-based, or likelihood-free, inference [8], and of amortized filtering, smoothing, and data assimilation; we claim no novelty there. The closest neighbors learn the dynamics surrogate itself: amortized assimilation recovers hidden states from noisy observations in chaotic systems with learned machinery [17], and likelihood-free

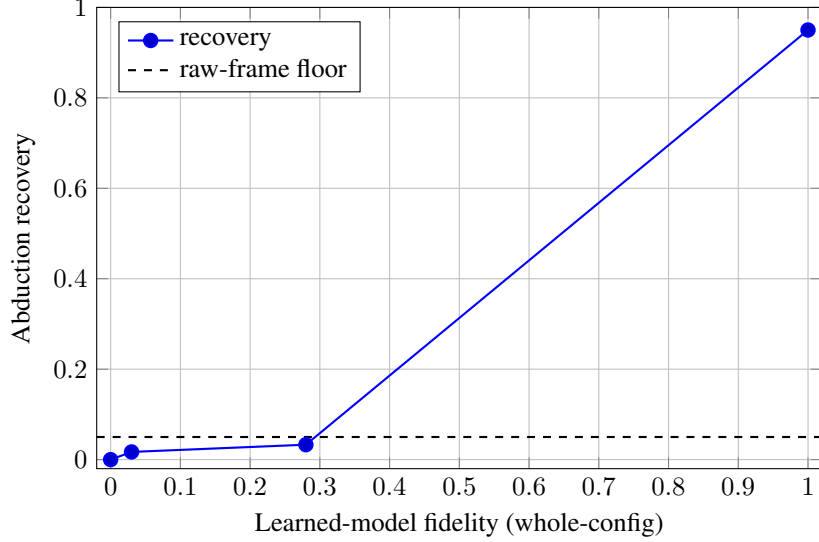


Figure 6: Degrade-to-chance: abduction recovery tracks the learned model’s fidelity and collapses to the floor when degraded. The dashed line marks the raw-frame floor at 0.05 (random-IC chance is $\approx 1.5 \times 10^{-5}$).

inference has been extended to state-space models whose transition dynamics are *unknown* and learned from data [1]. Our setting is a special, exact case of that family, with the candidate set enumerated rather than approximated and scoring done by exact match. What we add on top is not the inference machinery but the head-to-head against LLMs across the three axes and the dissociation it exposes.

The reversal curse and code-invertibility. That LLMs are weaker backward than forward is documented by the reversal curse [2] and by code-invertibility collapse [16], and o3 scoring near zero on long-horizon abduction is *consistent with* that directional bias. That this bias is at least partly a property of the training objective rather than a metaphysical limit is shown by reverse-training remedies that substantially improve bidirectional recall [11]; we accordingly frame our gap as a property of *current unaided models as evaluated*, not an architectural impossibility. Our contribution is the control-matched, cross-domain dissociation that survives a dedicated reasoning model, not the bare backward failure.

Abduction, inference modes, and causal benchmarks. A growing line treats abduction as a distinct inferential mode in LLMs: inductive, abductive, and deductive inference have been analyzed separately, with their relative difficulty found to depend on task and format [23], and dedicated abductive benchmarks now exist for event causal inference [4] and for structured neuro-symbolic abductive tasks, where adding explicit state and symbolic control improves performance [15]. A parallel causal-reasoning line reports that frontier models stay brittle on counterfactual and structured causal queries [13, 7]. We position against this ecosystem rather than claim abduction is unstudied: our operationalization is deliberately specific, *simulation-grounded exact initial-state recovery under known observation noise*, not natural-language event-cause selection or the full structural-causal-model counterfactual pipeline. The contribution is the benchmark construction and the learned-simulator-versus-unaided-LLM dissociation, not the existence of abductive or inverse inference as a research area.

Positioning against the nearest framing paper. The code-invertibility work [16] owns forward-vs-inverse asymmetry as a phenomenon. We name the three gaps we fill. First, it has no non-LLM *learned* simulator as the inverter. Second, it has no control axis; we add the matched within-domain control comparison that makes this a *dissociation* rather than just an asymmetry. Third, it is single-domain; we show the effect unchanged across a reversible 1-D CA and an irreversible 2-D world. We also note the Kaggle Conway’s-Reverse-Game-of-Life line [14], which uses our exact roll-

forward candidate-search MAP structure but with the *known* GoL rule, no LLM comparison, and no forward/control-vs-abduction dissociation. On wording and scope: AutomataGPT and related work [3] show that transformers *can* succeed at CA forward forecasting and ruleset inference, and own “inverse CA” in the sense of *rule* recovery. Our claim is narrower and orthogonal, exact *initial-condition* (hidden-state) recovery from a *noisy* trajectory; we always mean initial-condition recovery, never bare “inverse CA,” and we do not claim transformers cannot learn CA dynamics or rules.

Relation to model-based RL. Planning inside a learned model is the move of MuZero [19] and DreamerV3 [12]. Our reasoner shares the “simulate the learned model” stance, but it targets the *inverse* (abduction) rather than forward control, and it learns the rule from black-box transitions rather than reward-driven latents. Sequence-modeling of automata [3] and the broader “LLMs and structured reasoning” setting [20] are the adjacent threads.

Planning and control benchmarks. Our control axis sits in the established planning-evaluation line: PlanBench shows that ordinary LLMs plan poorly [21], while a follow-up finds that o1-style reasoning models improve substantially on it without saturating [22]. That trajectory is precisely our control result, a reasoning model closes much of the control gap, and it is why control is the right *matched* axis: the same models that close planning still do not close abduction.

Logic-augmented LLMs. Pipelines that offload formal reasoning to a solver via an LLM front-end, such as Logic-LM [18], share our division of labour (language as I/O, a non-LLM engine as the reasoner), but they target deductive and planning tasks, not initial-condition abduction over a black-box-learned simulator.

8 Limitations

The CA is multi-seed with CIs (ours $n = 200/\text{rule}$; LLMs $n = 15/\text{rule}/\text{horizon}$). The CA CIs vary the *evaluation instances*; to check the ours-equals-oracle claim is not an artifact of the training run, a training-seed sweep retrains the rule under five (*model-init*, *training-data*) seeds and scores abduction on a single fixed evaluation set ($H=12$, $n=60$): ours equals the oracle at *every* seed (56/60 Rule 110, 47/60 Rule 90, std = 0.000 across seeds), so the claim is robust to the training run (experiments/ca_seed_sweep.py). Sokoban is now $n = 36/\text{size}$ across $\{6, 7, 8\}$ with CIs (ours equals oracle), with the LLM contrast a bounded slice ($n = 8$, wide CIs, so the trend rather than a tight LLM CI is the point). The domains are abstract vision-free toys scored by exact equality. MAP enumeration is feasible only at small state spaces, and the bound is now measured: the CA has 2^{2L} candidates (so $L \lesssim 12$), and Sokoban candidates exceed the 20,000 cap past about 8×8 ($10 \times 10 = 97,527$), so scaling the inverse to larger boards needs amortized inference, which we do not build here. On Sokoban, A* also solves the *control* task, so the claim concerns the abduction *mode*, not a win over classical planning. We also state the inductive-bias asymmetry plainly: our inverter is a *hand-structured* simulator family (a weight-shared local rule matched to each substrate, with no width parameter and the exact recurrency form, plus exhaustive candidate search), compared against an LLM *forbidden from writing or calling code*. This is not general-reasoner-versus-general-reasoner; the precise reading is that an LLM struggles to do in-context exact inverse search over small noisy dynamical systems where a structured learned simulator plus exhaustive search succeeds up to the tractability wall, and the result is a guide for *when to reach past the language model and simulate*, not a claim that the simulator generalizes as a reasoner. o3 is not weak in general. It does forward and control fine and *partially* abducts, and on the CA it solves a few *short* trajectories. The narrow claim is that it does not *reliably* abduct and decays toward the floor as difficulty (trajectory length) rises. Finally, the reasoning-model evidence now spans *two* dedicated reasoners from different labs: o3 ($n = 15/\text{cell}$) and Claude Opus 4.8 in adaptive thinking ($n = 5/\text{cell}$, a smaller probe). Both close forward and control yet fail to abduct reliably while the gap to ours grows with horizon. The replication is partial in two senses we state plainly: Opus’s probe is small, and it recovers a larger fraction than o3 (so the dissociation is weaker for it). The claim is about *unaided* inference: with code execution the same model closes the gap (Section 5), so we do not claim LLMs cannot abduct with tools; only that, *unaided*, they cannot carry the exact inverse search in-head. We state the load on this evidence plainly: the tool-augmented result that *localizes* the failure to in-context search is one model (Opus 4.8) on one $n = 30$ slice, so it justifies the *unaided* qualifier rather than establishing the localization across models. The program-aided-reasoning literature [10, 6] makes the mechanism

general, but confirming it with a second tool-enabled reasoner (o3 with code execution) is future work. The title’s *unaided* claim does not depend on that single slice: it rests on the *unaided failures*, which are replicated across o3, Sonnet, and Opus. Strengthening the class claim further would add more reasoners (DeepSeek-R1, Gemini-2.5 thinking, o1), each both *unaided* and with code execution, and a larger n for the Opus probe.

9 Conclusion

We isolated abduction, the recovery of an initial condition from a noisy trajectory, as a distinct axis of reasoning, and showed a forward/control-pass, inverse-fail dissociation in *unaided* frontier LLMs. It survives two dedicated reasoning models from different labs (o3 and, in a smaller probe, Claude Opus 4.8) and holds unchanged across a reversible 1-D CA and an irreversible 2-D Sokoban. The inverter is a non-transformer reasoner whose forward rule is learned from black-box transitions; it sits at the true-dynamics recoverability oracle, and load-bearing controls show the learned model carries the inference, in a graded manner and inside the model. We cede the method to SBI [8], “LLMs fail backward” to the reversal-curse literature [2], and the inverted-wall principle to data assimilation in unstable subspaces [5].

The result has two practical readings. For evaluation, a scalar reasoning score earned on forward tasks does not certify a model for inverse ones: a model that plans well can still fail to reconstruct, so the abduction axis is worth measuring on its own. For building systems, wherever the job is to infer a hidden cause from its noisy consequences, the answer has to be checkable, and you either cannot grant arbitrary code execution or need guarantees over a best-effort program. There, a learned-simulation reasoner that inverts by MAP is a verifiable alternative to a frontier LLM, one that generalizes across system size by construction, abducting at widths it never saw in training. Naming the axis tells you when to reach past the language model and simulate.

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A LLM prompt and answer extraction

Every LLM is given the task with the rule made explicit; nothing about the dynamics is withheld. The abduction (inverse) prompt, modulo the instance, is:

```
A 1-D second-order cellular automaton on a ring of  $L$  cells uses
Wolfram rule  $R$  (table: the eight 3-neighbourhood outputs). The
dynamics are  $x[t+1][i] = f(x[t][i-1], x[t][i], x[t][i+1]) \text{ XOR } x[t-1][i]$ 
(indices wrap mod  $L$ ). Below are  $H+1$  consecutive rows
of one trajectory, but each bit was independently flipped with
probability 0.2 (observation noise). [the rows]. The whole
trajectory is determined by its first two rows via the rule. Using
the dynamics to denoise, recover the TRUE first two rows. Reason
step by step, then end with two lines EXACTLY: FINAL_X0: <L-bit
row> / FINAL_X1: <L-bit row>. Do not use any tools or code;
reason it out yourself.
```

The forward and control prompts share the template (the full rule table, the dynamics equation, and a strict final-answer tag FINAL: or FLIPS:). Extraction requires the exact tag; an output without it is recorded as a parse failure and reported separately, never as a reasoning failure. At $H = 18$, o3 produced a well-formed answer on 27/30 instances, so its 4/30 recovery (at high effort) is a reasoning result, not a formatting or truncation artifact.

What the LLMs do. Their visible reasoning shows the failure is genuine. Claude Sonnet correctly identifies the MAP procedure but cannot execute it by hand: “I’ll try to find $x[0]$ and $x[1]$ by testing candidates ... minimizing total bit flips ... there are $256 \times 256 = 65536$ possibilities.” It names exactly our estimator, then must enumerate it in its head and fails. A representative o3 attempt ($H = 18$, Rule 110) concludes “the only initial pair that keeps every later computed row in the closest overall agreement ($\leq 20\%$ error for every row) is FINAL_X0: 00110001 ...”: o3 settles for an approximate reconstruction, but abduction is scored on exact recovery, so it misses.

B The inverted wall

One further observation concerns the dynamics themselves, separate from the dissociation, which is why it is kept out of the main results. On *true* dynamics (training-free), the likelihood gap separating the true IC from the best alternative is wider, and still growing, under irreducible Rule 110 than under reducible Rule 90. Computational irreducibility *aids* the inverse, the reverse of the forward intuition. Figure 7 plots the per-bit likelihood gap (mean $\pm$ std over 5 seeds, noise 0.2). At $H = 8$, Rule 110 is 0.019 ± 0.004 and Rule 90 is 0.008 ± 0.003 ; at $H = 12$, 0.051 ± 0.003 vs 0.021 ± 0.002 ; at $H = 16$, 0.075 ± 0.002 vs 0.030 ± 0.002 ; at $H = 20$, 0.095 ± 0.001 vs 0.036 ± 0.001 . The ordering Rule 110 $>$ Rule 90 holds at every H on the *worst* of 5 seeds. Sensitive dependence makes each chaotically-decorrelated frame a fresh measurement, which sharpens identifiability, while the additive rule’s low-growth difference-mode keeps a bounded gap. We lead with the likelihood gap because the per-bit error ordering is *not* seed-robust.

We cede the *principle*, the known sensitivity $\leftrightarrow$ identifiability relation, or data assimilation in unstable subspaces [5], and keep only the class-4-vs-class-2 empirical instantiation, which Elser [9] flags as open (the CA-class $\rightarrow$ reconstruction-hardness translation). The abduction-dissociation headline does not depend on it.

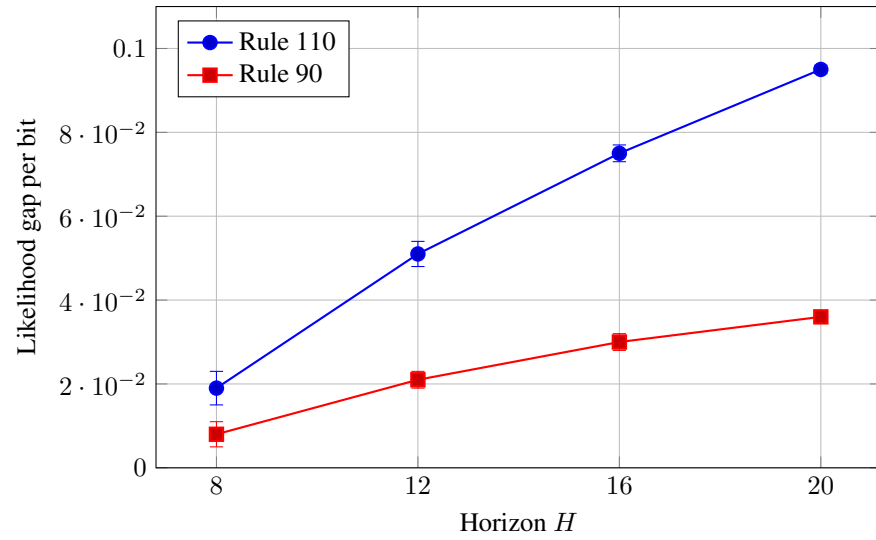


Figure 7: The inverted wall: irreducible Rule 110 opens a wider, growing identifiability gap than reducible Rule 90 (the ordering holds on the worst of 5 seeds). Mean $\pm$ std over 5 seeds; observation noise 0.2.